AI-Driven Cybersecurity Threat Detection Platform

Purpose of this Document
This PDF consolidates all daily learning, technical concepts, and interview-relevant explanations from the complete development of the AI-Driven Cybersecurity Threat Detection project. The focus is on conceptual clarity with real-life examples rather than memorization.

Day 1 – Problem Understanding & Architecture

Cybersecurity systems must identify suspicious user behavior early. The goal was to design an AI system that classifies user activity as NORMAL or THREAT.

Architecture Overview:
User → Web UI → Flask API → ML Model → Prediction → Response.
This mirrors real enterprise SOC (Security Operations Center) pipelines.

Day 2 – Data Design & Feature Engineering

We designed behavioral features such as login failures, IP attempts, session time, and admin access. These features simulate logs generated in real systems like firewalls, SIEM tools, and authentication servers.

Example: Multiple login failures in short time often indicate brute-force attacks.

Day 3 – Data Preprocessing

Raw data cannot be directly used by ML models. We validated schema, handled numeric consistency, and applied scaling using StandardScaler.

Real-life analogy: Just like medical tests must be normalized before diagnosis.

Day 4 – Model Training

We used an anomaly detection model suitable for cybersecurity where labeled attack data is rare. The model learned normal behavior patterns and flagged deviations.

Example: If a user suddenly accesses admin resources at odd hours, it is treated as anomalous.

Day 5 – Model Evaluation

The model produced a decision score. Scores below threshold were classified as THREAT. This mimics enterprise risk scoring systems used by banks and security firms.

Day 6 – Backend API Development

Flask API exposed the model using REST endpoints. JSON input simulated real-time log ingestion from applications.

Example: A login microservice sending behavior metrics to a central security engine.

Day 7 – Frontend, Integration & Deployment

A simple UI allowed non-technical users to interact with the AI system. Deployment on Render made the system accessible globally.

This is identical to real SaaS security dashboards used by companies.

End-to-End Project Working (Real-Life Explanation)

Imagine an online banking system. A user logs in repeatedly with wrong passwords, tries multiple IP addresses, and suddenly requests admin-level access. These events are captured as logs. Our AI system processes these logs, normalizes the data, evaluates behavior against learned normal patterns, and flags the activity as a THREAT in real time.

This is how modern AI-based cybersecurity systems work in enterprises like banks, cloud providers, and fintech platforms.

Key Interview Takeaway
This project demonstrates not just machine learning, but system design, security thinking, API development, cloud deployment, and AI lifecycle management — core responsibilities of an AI Architect.